

Image Optimization Guide for EPUB

MANDATORY Image Requirements for Bestseller Quality

1. Decorative Brushstroke & SVG Image Optimization

Brushstroke Background (brushstroke.jpeg)

The brushstroke used throughout this book is a raster image in JPEG format rather than an SVG. To keep the file size reasonable and maintain visual quality, export the brushstroke as a high-quality JPEG:

- Save the file as `brushstroke.jpeg` in the `OEBPS/images/` folder.
- Use 85 % quality with progressive encoding for faster loading.
- Ensure the final file size is below **150 KB**.

Optimizing Other SVG Assets

While the brushstroke itself is a JPEG, other small icons or decorative elements may still be stored as SVG files (e.g., quiz icons or worksheet icons). To optimize these SVGs:

```
<!-- Example SVG optimization: remove unnecessary metadata and simplify paths -->
<svg width="150" height="150" viewBox="0 0 150 150" xmlns="http://www.w3.org/2000/svg">
  <path d="M20 75 Q30 45 60 35 Q90 30 120 45 Q135 55 130 75 Q125 95 100 110 Q70
120 40 105 15 Q15 85 20 75 Z"
      fill="#4ECDC4" opacity="0.9"/>
</svg>
```

SVG Optimization Checklist: - ☐ Remove unnecessary metadata and comments - ☐ Minimize decimal places in path data - ☐ Use efficient gradients and filters - ☐ Compress without quality loss - ☐ File size target: Under **5 KB** per SVG

2. Raster Image Optimization

JPEG Images (portraits, photographs) - **Quality:** 85% (optimal balance of quality/file size) - **Encoding:** Progressive JPEG for faster perceived loading - **Color Profile:** sRGB for consistent color reproduction - **Maximum Width:** 1200px for full-width images - **File Size Target:** Under 150KB per image

PNG Images (graphics, illustrations) - **Bit Depth:** 8-bit when possible, 24-bit for complex images - **Optimization:** Use PNG optimization tools (OptiPNG, PNGCrush) - **Transparency:** Optimize alpha channel efficiency - **File Size Target:** Under 100KB per image

3. Responsive Image Implementation

Standard Implementation:

```

```

Accessibility Enhanced:

```
<figure class="image-container" role="img" aria-labelledby="img-caption-1">
  
  <figcaption id="img-caption-1" class="image-caption">
    Sarah's transformation: From corporate conservative to confident and
stylish
  </figcaption>
</figure>
```

4. Performance Optimization Techniques

Lazy Loading Implementation:

```
<!-- For non-critical images -->


<!-- For critical images (above the fold) -->

```

CSS for Responsive Images:

```
.content-image {
  max-width: 100%;
  height: auto;
  display: block;
  margin: 1rem auto;
  border-radius: 8px;
  box-shadow: 0 4px 8px rgba(0,0,0,0.1);
}

.brushstroke-img {
  position: absolute;
  width: 100%;
  height: 100%;
  object-fit: cover;
  z-index: 1;
}

.closing-image {
  max-width: 200px;
  height: auto;
  display: block;
  margin: 2rem auto;
  opacity: 0.8;
}
```

5. Image Format Decision Tree

Use SVG for: - Icons and simple graphics - Scalable decorative elements such as icons or simple line graphics - Line art and logos - Mathematical diagrams

Use JPEG for: - Photographs of people - Complex scenes with many colors - Hair styling examples - Before/after comparisons

Use PNG for: - Graphics with transparency - Simple illustrations - Text overlays - Screenshots with text

6. Accessibility Requirements

Alt Text Guidelines:

Decorative Images:

```

```

Informative Images:

```

```

Complex Images:

```

```

7. File Naming Convention

Organized File Structure:

```
OEBPS/images/
├─ brushstroke.jpeg (chapter decorations)
├─ closing-ornament.png (chapter endings)
├─ cover.jpg (book cover)
├─ author-photo.jpg (author information)
├─ chapter-01/
│   ├─ hero-image.jpg
│   ├─ technique-demo.jpg
│   └─ before-after.jpg
├─ chapter-02/
│   ├─ hero-image.jpg
│   └─ tools-layout.jpg
└─ icons/
    ├─ quiz-icon.svg
    └─ worksheet-icon.svg
```

File Naming Rules: - Use lowercase letters only - Separate words with hyphens - Include descriptive keywords - Keep names under 50 characters

8. Optimization Tools and Commands

Command Line Tools:

```
# JPEG optimization
jpegoptim --size=150k *.jpg

# PNG optimization
optipng -o7 *.png

# SVG optimization
svgo --multipass *.svg

# Batch resize images
mogrify -resize 1200x1200\> -quality 85 *.jpg
```

Online Optimization: - TinyPNG for PNG compression - JPEG.io for JPEG optimization - SVGOMG for SVG optimization

9. Quality Assurance Checklist

Before finalizing each image: - [] File size meets target requirements - [] Quality is appropriate for intended use - [] Alt text is descriptive and meaningful - [] Image displays correctly at various sizes - [] Color reproduction is accurate - [] Loading performance is optimized - [] Accessibility requirements met

10. EPUB-Specific Considerations

Device Compatibility: - Test on high-DPI displays (Retina, etc.) - Verify rendering on grayscale e-readers - Check performance on older devices - Ensure images work in night mode

File Size Limits: - Individual image: Max 1MB - Total EPUB size: Target under 50MB - Consider device storage limitations - Balance quality vs. file size

11. Implementation in Codex Commands

Image Processing Commands:

```
@agents OPTIMIZE ALL IMAGES for performance and accessibility:

1. IMAGE OPTIMIZATION:
  - **Brushstroke JPEG:** Save `brushstroke.jpeg` at 85 % quality with
    progressive encoding and keep the file size under 150 KB.
```

- ****Other JPEGs:**** Apply 85 % quality and progressive encoding; ensure a maximum width of 1 200 px; compress to meet size targets.
- ****PNG files:**** Use 8-bit when possible and optimize transparency.
- ****SVG icons:**** Remove unnecessary metadata and comments, simplify paths and compress each SVG to under 5 KB.

2. ACCESSIBILITY IMPLEMENTATION:

- Add descriptive alt text to every image
- Use semantic `<figure>`/`<figcaption>` markup where appropriate
- Implement proper ARIA labels
- Ensure screen reader compatibility

3. RESPONSIVE IMPLEMENTATION:

- Add CSS `for` responsive sizing
- Include lazy loading attributes
- Optimize `for` various screen sizes
- Test cross-device compatibility

This comprehensive optimization ensures your EPUB delivers professional quality while maintaining excellent performance across all devices.
